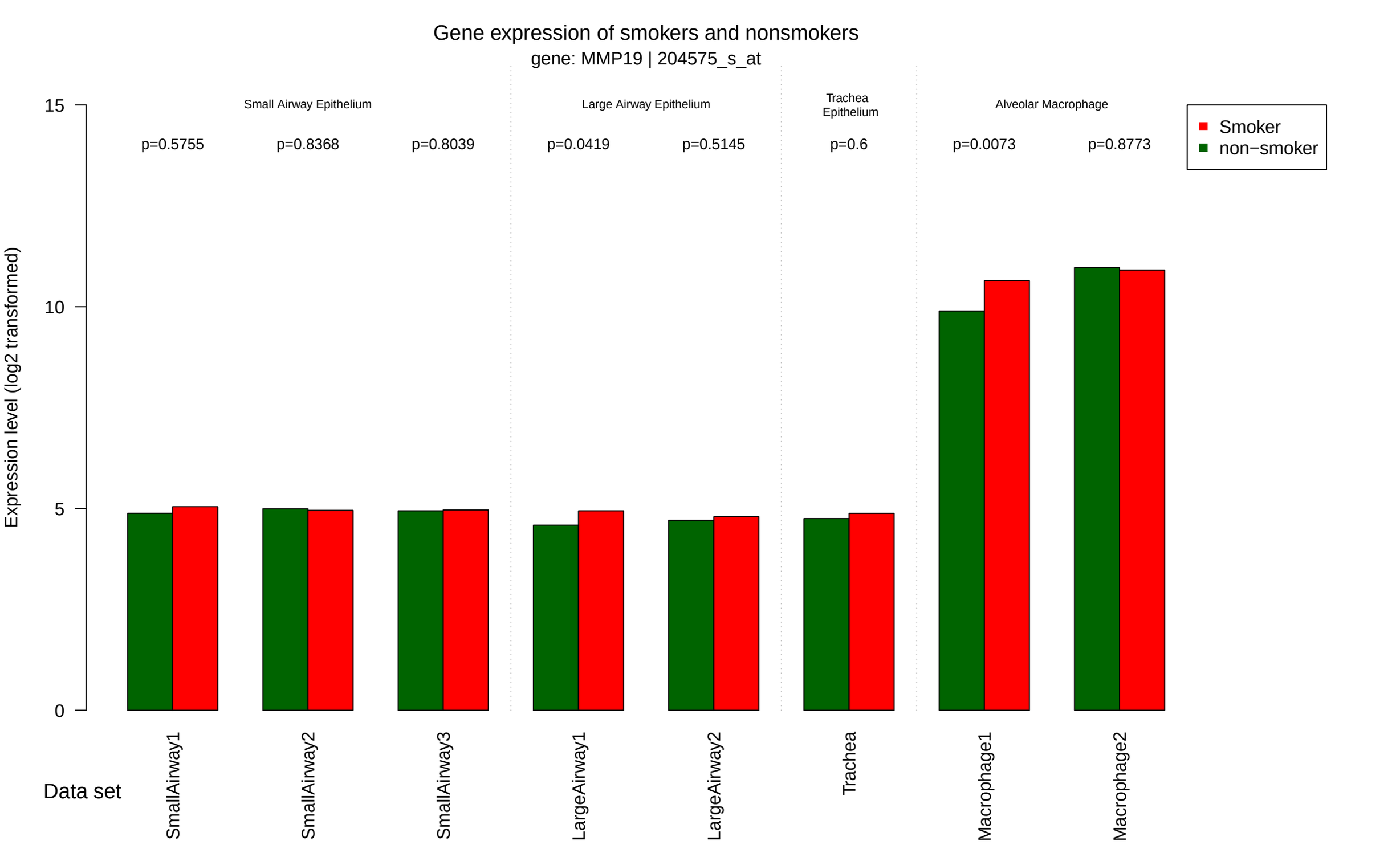
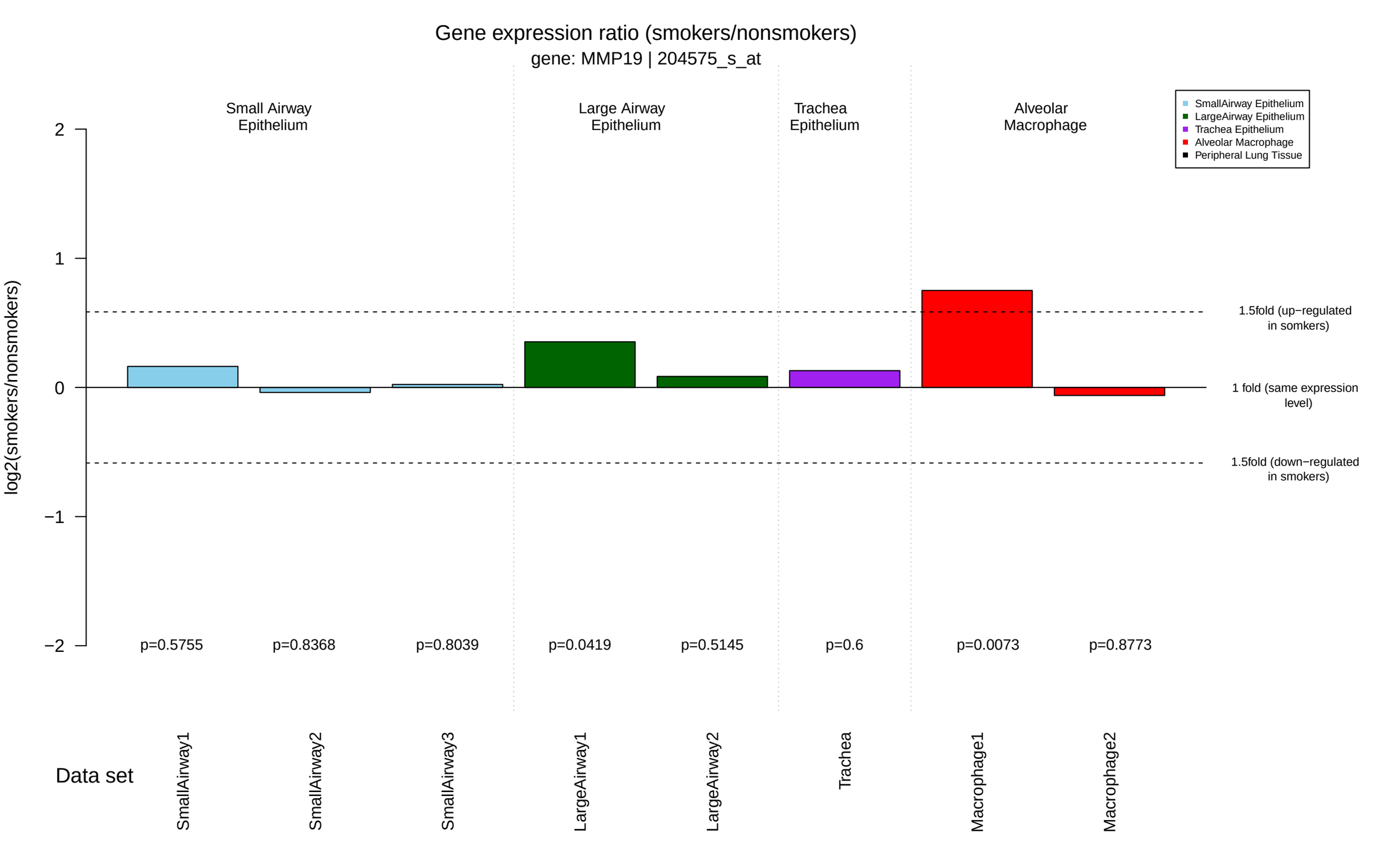
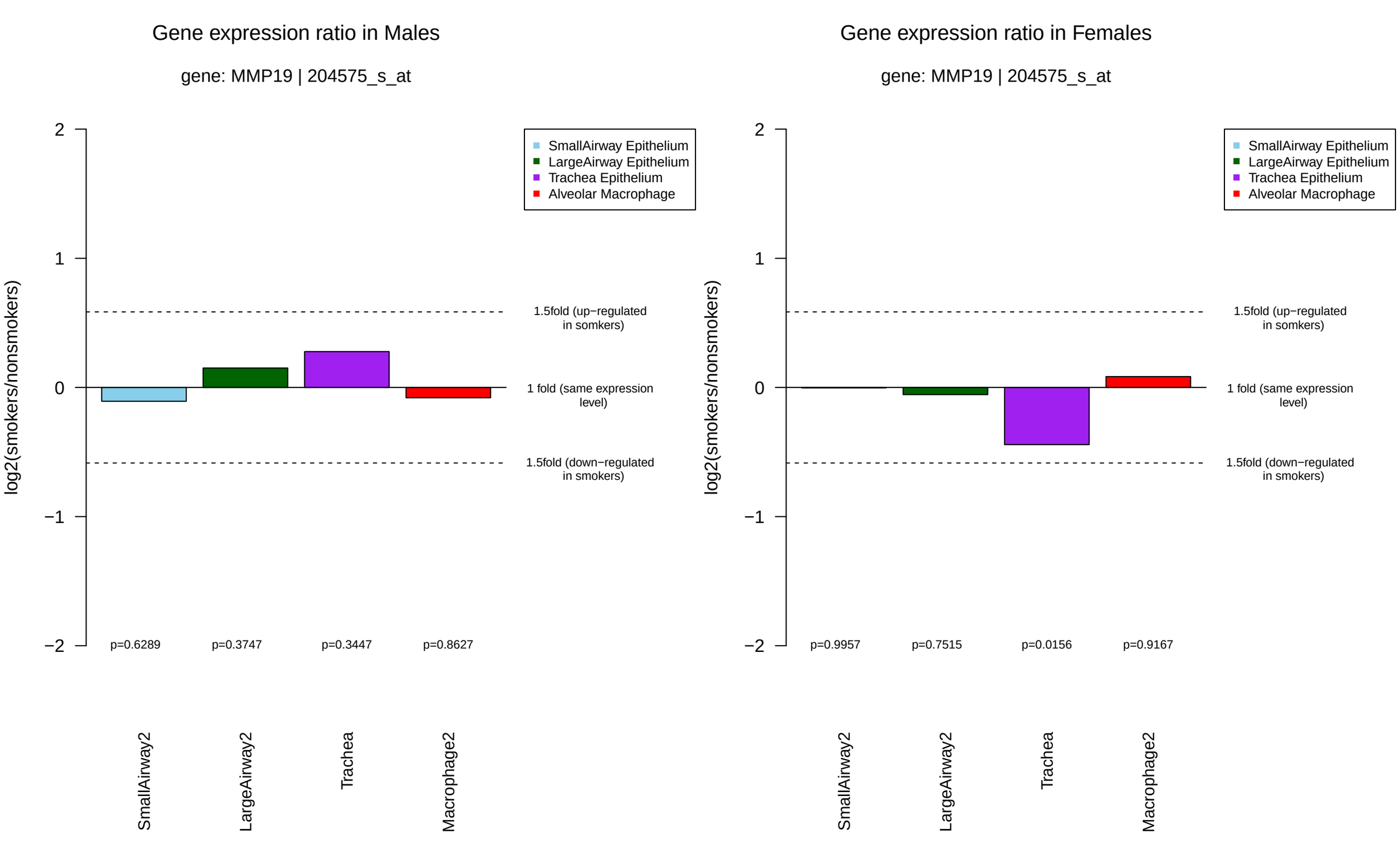
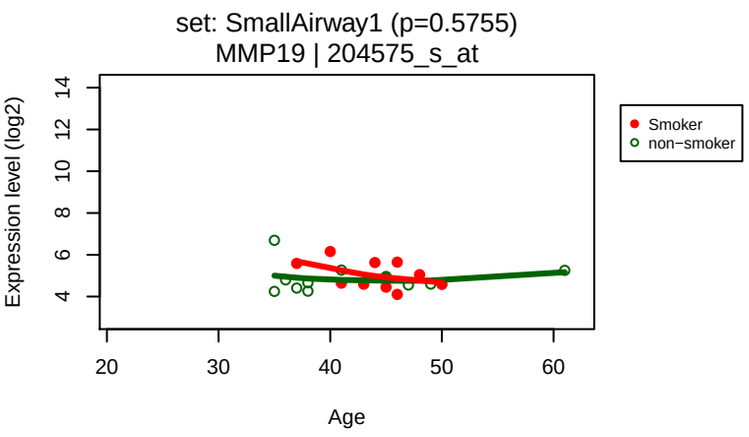
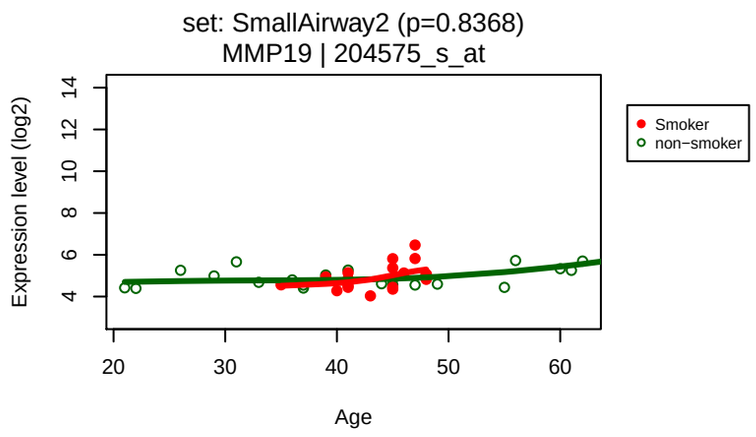
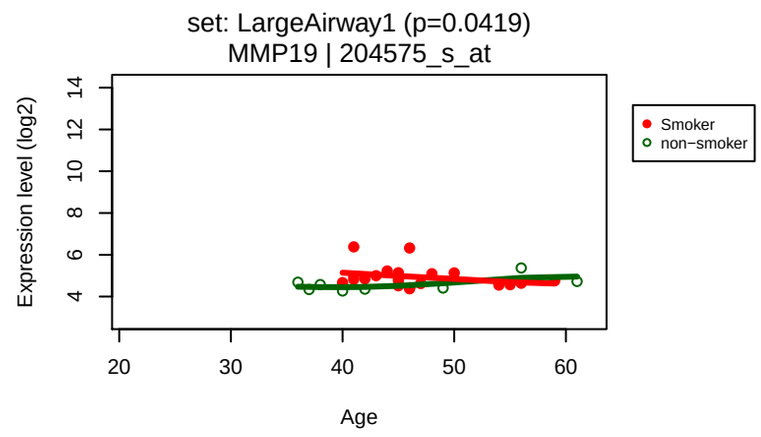

Gene expression of smokers and nonsmokers
gene: MMP19 | 204575_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5755
p=0.8368
p=0.8039
p=0.0419
p=0.5145
p=0.6
p=0.0073
p=0.8773
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

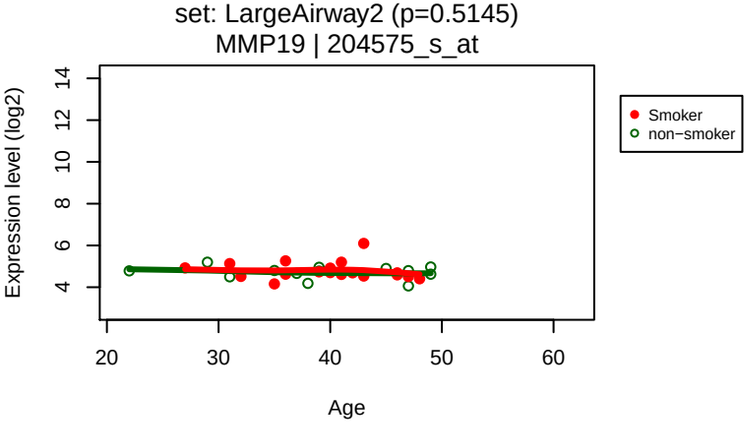

Gene expression ratio (smokers/nonsmokers)
gene: MMP19 | 204575_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5755
p=0.8368
p=0.8039
p=0.0419
p=0.5145
p=0.6
p=0.0073
p=0.8773
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

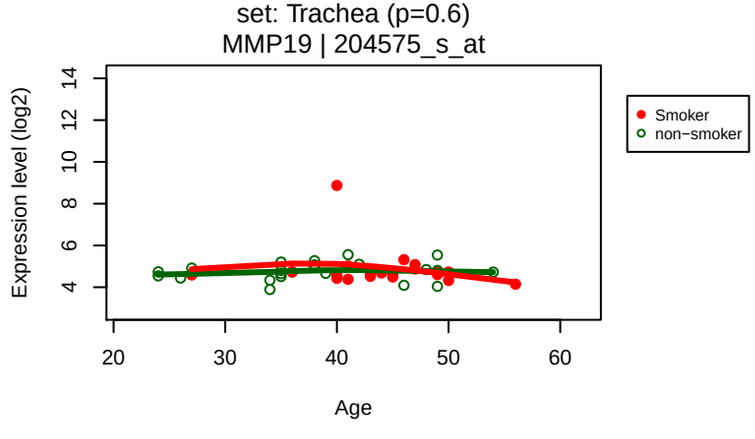

Gene expression ratio in Males
gene: MMP19 | 204575_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6289
p=0.3747
p=0.3447
p=0.8627
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP19 | 204575_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9957
p=0.7515
p=0.0156
p=0.9167
SmallAirway2
LargeAirway2
Trachea
Macrophage2

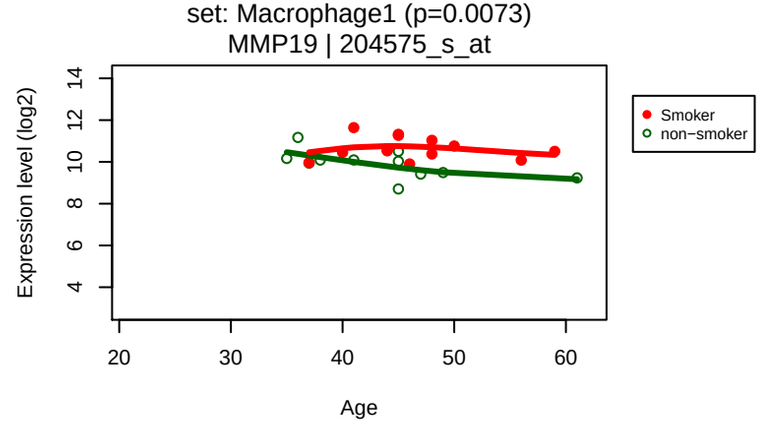

set: SmallAirway1 (p=0.5755)
MMP19 | 204575_s_at
Expression level (log2)
Age
Smoker
non−smoker

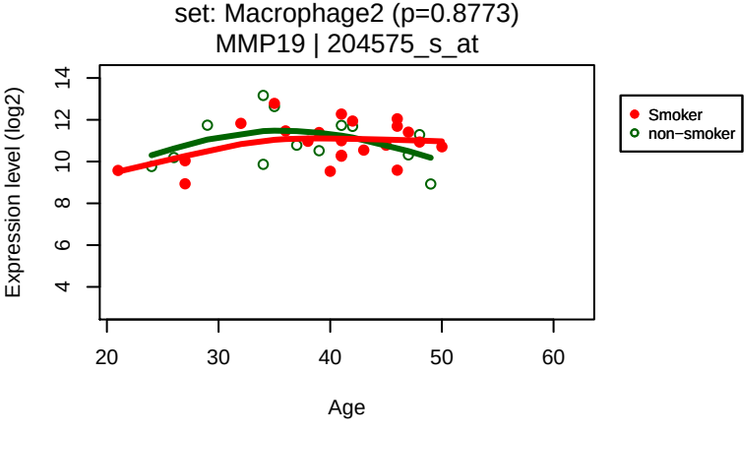

set: SmallAirway2 (p=0.8368)
MMP19 | 204575_s_at
Expression level (log2)
Age
Smoker
non−smoker

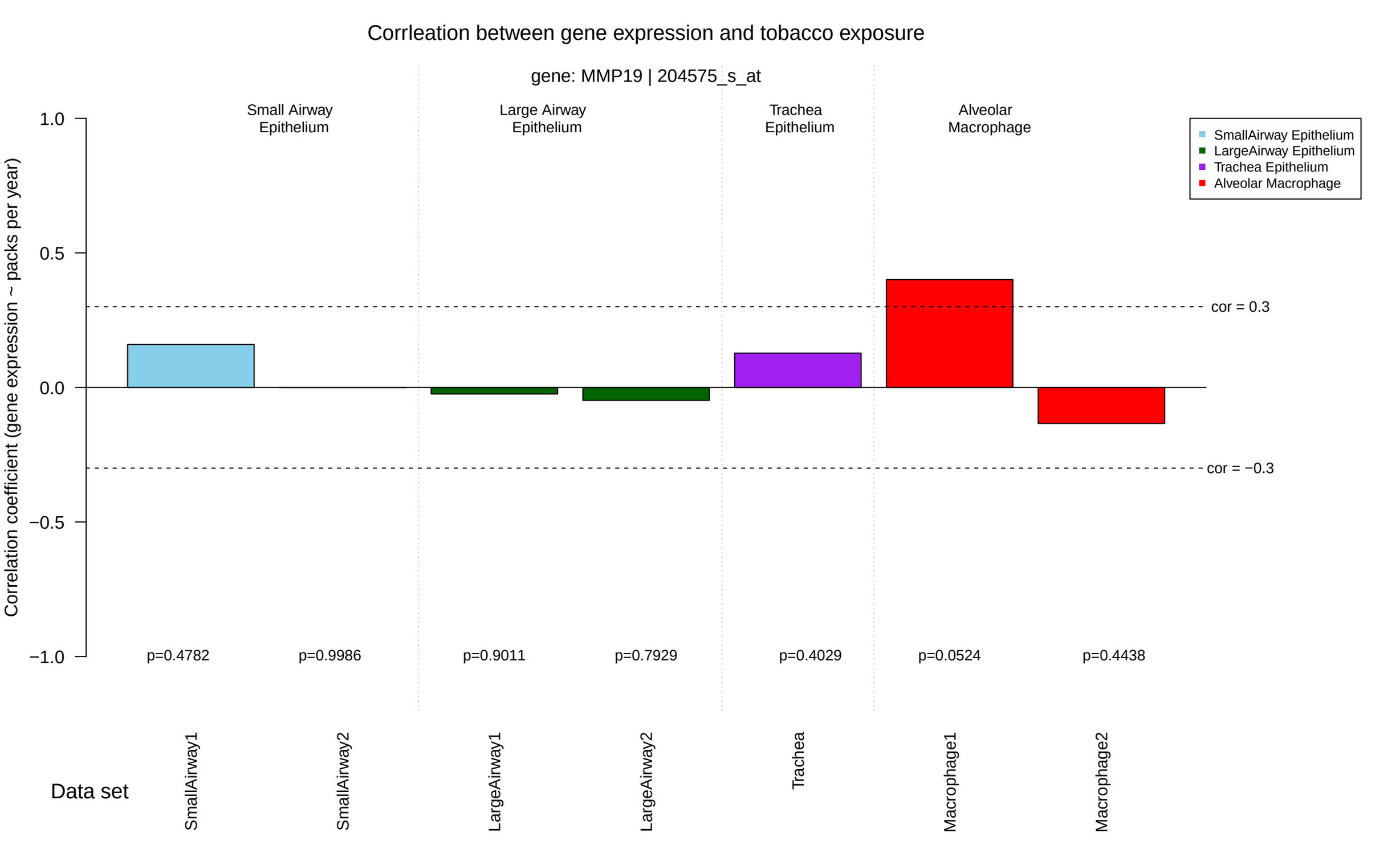

set: LargeAirway1 (p=0.0419)
MMP19 | 204575_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5145)
MMP19 | 204575_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6)
MMP19 | 204575_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0073)
MMP19 | 204575_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8773)
MMP19 | 204575_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP19 | 204575_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4782
p=0.9986
p=0.9011
p=0.7929
p=0.4029
p=0.0524
p=0.4438
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2